it as a store of value or to facilitate exchange. Anything that succeeds in gaining acceptance has the potential to become money or currency—from sea shells, pepper and gold to paper and digital. Digital, because the world is now a village, and with real-time global trade and exchanges between organisations, businesses, governments and now peer-to-peer growing exponentially, digital money is needed to facilitate these instantaneously. Digital money also serves an important purpose of compression of value (allowing for speedy movement and sometimes invisibility) that in the past art and gems have provided," says Rajesh Lalwani, CEO, Scenario Consulting Pvt Ltd, preparing the ground for a discussion on this essential financial instrument!

Digital currency is an umbrella term, which refers to currency that exists in electronic form. It is stored, managed, and transacted through computer systems, other electronic devices, and wallets. Digital currencies do not have physical attributes, like notes or coins—though they may be equivalent to or convertible to fiat currency. There is no minting, storage, transportation, or other logistical processes associated with digital currency. Digital currency transactions are inexpensive and almost instantaneous. They enable seamless transfer of value—even across borders.

If the system is based on distributed ledger technology (DLT), each transaction is linked to the previous one and the next one in a distributed ledger. So, to change one transaction, you will have to change the entire chain. This ensures that all transactions are traceable but not reversible, making them very difficult to tamper with. All transactions are encrypted between digital wallets, resulting in exact parity calculation on the ledger. This instils a sense of trust in the system, reduces the possibility of errors, and the need for extensive reconciliation processes.

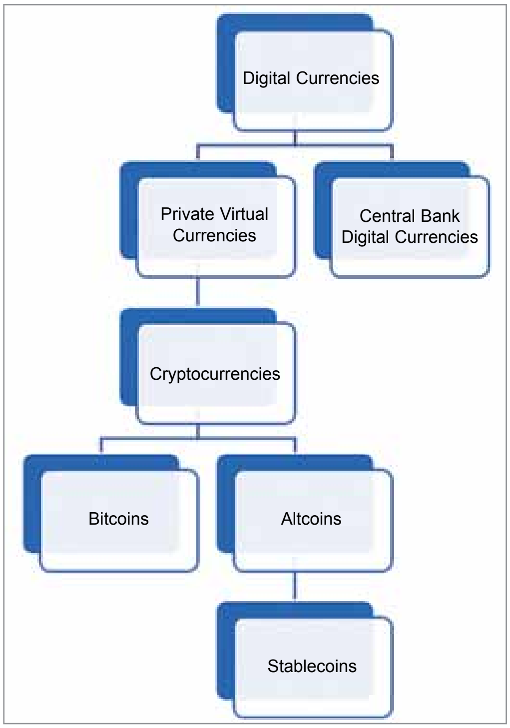

Fig. 1: Types of digital currencies

Given all the advantages, we have to remember that even the most robust and secure digital systems have fallen prey to hackers at some point of time. There have been rare cases of hackers stealing from wallets, or tampering with the protocol and making the currency unusable!

Overall, the concept of digital currency is definitely a desirable one, which can transform the world of finance. However, not all digital currencies are the same—and some can be very problematic too.

The many types of digital currency

Digital currencies can be either open or closed, regulated or unregulated, centralised or decentralised—which gives rise to different categories: private virtual currencies, cryptocurrencies, and central bank digital currencies. We must first be able to differentiate between these, to discern which is better or worse.

Private virtual currency is mostly unregulated. It is issued and controlled by a private issuer, such as the developer or founding organisation of a platform, and not by a central bank. It is usually native to a network, and has little value outside of it. It can be algorithmically controlled. Transactions are usually direct, party to party, without intermediaries. It may be centralised or decentralised, may or may not be encrypted, and may or may not be convertible to fiat currency.

Cryptocurrency is a form of private virtual currency, which uses cryptography to manage and control the creation of currency, as well as to secure and verify transactions. This makes it difficult to counterfeit or double-spend cryptocurrencies. Cryptocurrencies are decentralised virtual currencies, which depend on blockchain networks to validate transactions and instil trust in the system. The most popular cryptocurrency is the Bitcoin, which was developed in 2008 by an unknown person or group by the name of Satoshi Nakamoto. Bitcoins are basically small pieces of code, given to reward mining—a compute-intensive process that verifies Bitcoin transactions on the blockchain.

Altcoins refer to all cryptocurrencies other than Bitcoin, such as Ether, Dogecoin, etc. These use cryptography too, but may use different logic to validate transactions or include advanced features to solve the inherent problems of Bitcoin such as volatility.

Stablecoins are a type of altcoin, which peg the value of the cryptocurrency to some physical entity like fiat currency, gems, or precious